

Density Functional Theory

September 9, 2026

1 Introduction

In this project we will simulate, study and analyse the structure of a small cluster of 5 silicon atoms, Si_5 , thanks to Quantum Espresso [1, 2]. The workflow will start with a blind search of some possible equilibrium structures of Si_5 with a simple 'relax'. Then we will study the optimization of the energy cut-off and the cell size. With this preliminary study it will be possible to proceed with a deeper study of the structure, between the ones found previously, with the lowest energy, which is supposed to be also the most stable. Finally we will perform a more rigorous search for the minimum energy structure with a Simulated Annealing schedule. It is possible to consult all the codes used at the GitHub page [3].

2 Methods

2.1 Initial Search

We are interested in searching for possible stable structures of this Si_5 cluster. To do so, we implement a QE simulation that, starting from random initial coordinates, relaxes the structures towards energy minima. The simulation identifies a minimum when convergence is reached for the ionic and electronic part. For the nuclei this happens if the energy and forces variations fall below a certain value: 'etot_conv_thr', that is set to $10^{-5} Ry$, and 'forc_conv_thr', set to $10^{-4} Ry$. For this atomic coordinate the algorithm chosen was 'BFGS'. The k grid is based on Γ point and the cell, although its size is yet to be studied, has been fixed to a cube of side $20\text{\AA}$. This cell should be large enough already but the actual convergence will be carried on later. Similarly the energy cut-off has been set to a generic $40 Ry$ for the plain wave and $320 Ry$ for the charge density, as generally required by ultrasoft pseudopotentials like 'Si.pbe-nl-rrkjus_psl.UPF', used in these simulations. Since the system is isolated, it has been treated with fixed occupations, while the electronic convergence threshold has been set to $10^{-10} Ry$ with a mixing factor of 0.7.

We start from 5 initial random configurations and analyse the minimum they converge to based on their final structure and energy. To do so we will study their energies relative to one another and the values of the distances between all the 10 unique couples we can describe with 5 atoms. This comparison and a look at their 3D image will give us the necessary confidence to state if two minima are the same or not. Finally, it is important to underline that this study cannot prove whether the lowest energy structure found is the actual absolute energy minimum for the Si_5 cluster.

2.2 Optimization

After reaching a preliminary set of possible minima we want to look for the best parameters to set, as said before, for the cell size and energy cut-off. To do so, for every one of the stable structures found previously, we will first make a Self Consistent Field (scf) calculation changing 'ecutwfc' from $30Ry$ to $60Ry$ with step of $5Ry$. 'ecutrho' has been always kept as 8 times the value of 'ecutwfc'. In this way we want to analyse if the structures remain stable and how their absolute and reciprocal energy change due to this parameter. If a value for the energy cut-off doesn't show appreciable changes in stability, total energy or hierarchy with respect to the $60Ry$ step, it will be considered valuable. Obviously the smallest one of these energy cut-offs will be the one that minimizes the computational costs while keeping the simulation coherent with the tightest cut-off. This will be the converged value. The threshold for the maximum difference in total energy with respect to the $60Ry$ step will be decided based on the magnitude of the energies that the system displays normally. After choosing the best energy cut-off in such a way, we can fix it and move towards an analogous study for the cell size. Since the cell is kept cubic the parameter is only one: the linear size. This is valued from $14\text{\AA}$ to $24\text{\AA}$ with a $2\text{\AA}$ step. The same reasoning for the energy will be used to discriminate the best cell size that compromises between computational costs and accuracy. Finally, after finding the converged values for these two parameters we can reiterate the same blind search done before to find the actual lowest energy minimum, between the ones found here.

2.3 Electron Density

Between the stable structures found previously with the new values of the converged parameters, we identified the one with minimum energy. Our goal is now to analyse its electron density. To do that we aim to use the visualizer XCrySden [4, 5]. For this study we need to start a new 'SCF' simulation with the same parameters and pseudopotential defined in Section 2.1, with the exception of the energy cut-off and cell size that will be set as the converged values found before. This will allow us to write a 'pp.x' code that creates a file, 'si5_rho.xsf', ready to be visualized with XCrySden [4, 5].

2.4 Annealing Search

Finally, we implement a better search of minima of the space of possible configurations thanks to a technique called Simulated Annealing. This time, instead of starting from a random structure and looking for the associated minimum simply following the decrease of the energy, we introduce temperature and, therefore, thermal fluctuations that require molecular dynamics. Starting from high temperatures we give the molecule enough energy to move in the phase space and maybe explore new configurations. Then, gradually lowering the system to room temperature (300 K), we will effectively freeze the so obtained structure. The high temperature and stochastic fluctuations, at the beginning, allow to avoid getting stuck on relative minima. After the annealing schedule has finished the structure provided is relaxed again in order to deprive it from the spurious residual kinetic energy. However, this study is not sufficient, as before, to confirm that the absolute minimum has been found.

The procedure adopted consists in starting the system at 1800 K and lowering it, quite abruptly, to 300 K in a total of 150 steps, i.e. lowering the temperature of 150 K every 15 steps. The timestep has been set to 20 Rydberg atomic units. This quite rapid schedule has been designed to compromise with the computational costs. Also for this reason the convergence threshold for the electronic part has been increased to $10^{-5}Ry$ and the mixing factor is brought to 0.1. The parameters converged before (energy cut-off and cell size) will be fixed to the values found by the optimization.

3 Results

3.1 Initial Search

With this scouting we were able to find 3 stable configurations, since 3 of the initial ones fell into the same equilibrium point. Energies are in Table 1

Structure	Minimum	Energy (Ry)	ΔE (Ry)
1	min1	-50.6576133	0.0000000
2	min2	-50.4839137	0.1736995
3	min1	-50.6576127	0.0000006
4	min1	-50.6576105	0.0000027
5	min5	-50.5738874	0.0837258

Table 1: Energies and relative energy differences of the identified structures.

The energy differences already suggest that the first, third and fourth structures coincide and looking at the relative distances between the 5 atoms this is confirmed. We can see that the minimum to which these 3 structures converged towards has also the lowest energy found in this small search.

Seen the magnitude of the energies for the minima found, the threshold for the acceptance of the energy cut-off or cell size will be set to $10^{-4} Ry$. Any value that is closer to the energy of the simulations with the biggest energy cut-off or cell linear size will be considered acceptable, as long as the system does not show other changes in stability or hierarchy.

3.2 Optimization

From the results in the previous small search we have 3 possible minima that we will use to study the convergence of energy cut-off and cell size. The results are show in Figure 1.

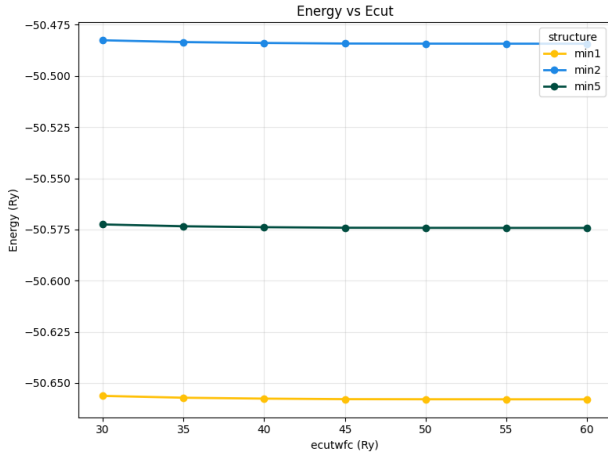
In these images it is visible that the first points that show no change in hierarchy and are closer than the threshold, $10^{-4} Ry$, to the energy of the last studied value, for all minima, have 'ecutwfc=45Ry' and cell 'linear size=16Å'. These will be the parameters chosen and with them fixed we will redo all the relaxes simulations done in the first search, starting from the same initial sets of coordinates.

minimum	ΔE (Ry)	ΔE (Ry)
	at ecutwfc=45Ry	at linear size=16Å
min1	$8.3 \cdot 10^{-5}$	$-2.6 \cdot 10^{-6}$
min2	$8.5 \cdot 10^{-5}$	$2.4 \cdot 10^{-6}$
min5	$9.0 \cdot 10^{-5}$	$1.3 \cdot 10^{-5}$

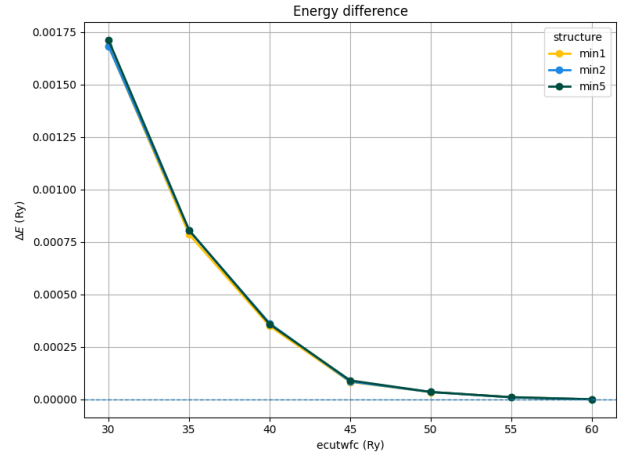
Table 2: Results of the convergence study for the values of 'ecutwfc' and the linear size of the cell for the chosen values

The results of this reiterate search are shown in Figures 4, in the Appendix, and in Table 3.

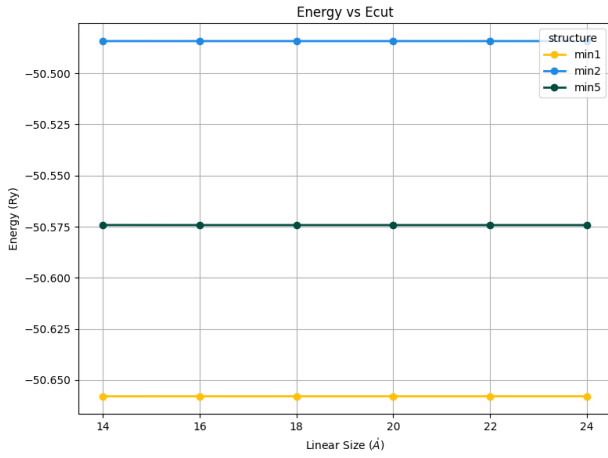
Applying the same observations as before we can conclude that the first, third and fourth simulations converged to the same structure. This statement is supported by Table 5 in the Appendix. This time the third one has the lowest energy of all.



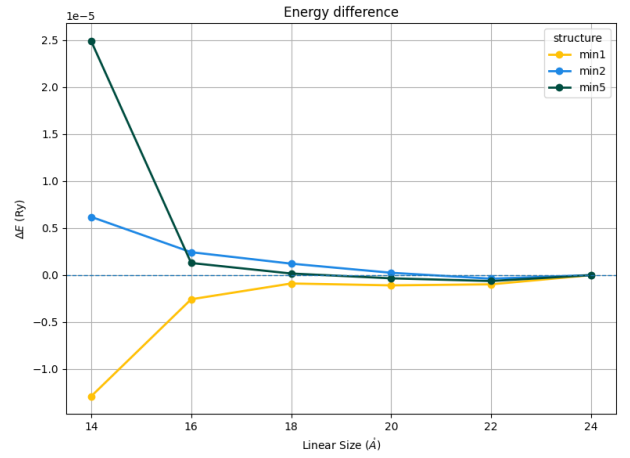
(a) Energy variation with the varying value of 'ecutwfc'



(b) Differences between the energy at 'ecutwfc=60Ry' and the current 'ecutwfc'



(c) Energy variation with the varying value of linear size of the cell



(d) Differences between the energy at 'linear size=24Å' and the current value of the linear size

Figure 1: Results of the convergence study for the values of 'ecutwfc' and the linear size of the cell

Structure	Minimum	Energy (Ry)	ΔE (Ry)
1	min1	-50.6579275	0.0000002
2	min2	-50.4842374	0.1736901
3	min1	-50.6579277	0.0000000
4	min1	-50.6579253	0.0000024
5	min5	-50.5742100	0.0837176

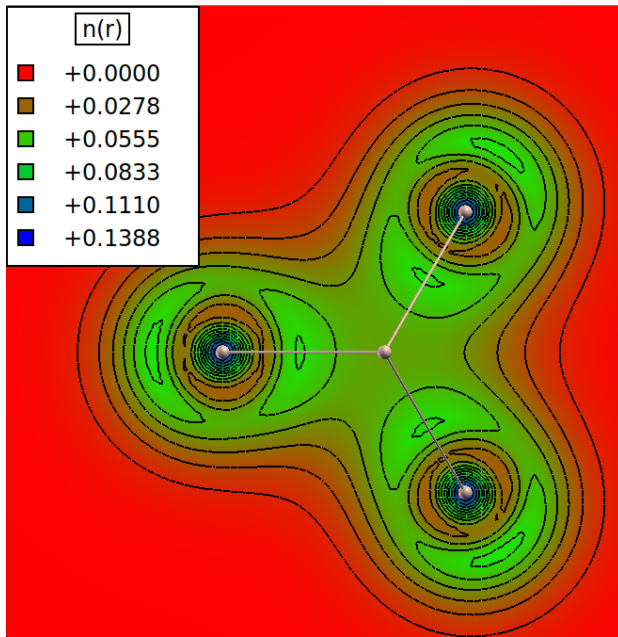
Table 3: Energies and relative energy differences of the reiterated relaxes.

3.3 Electron Density

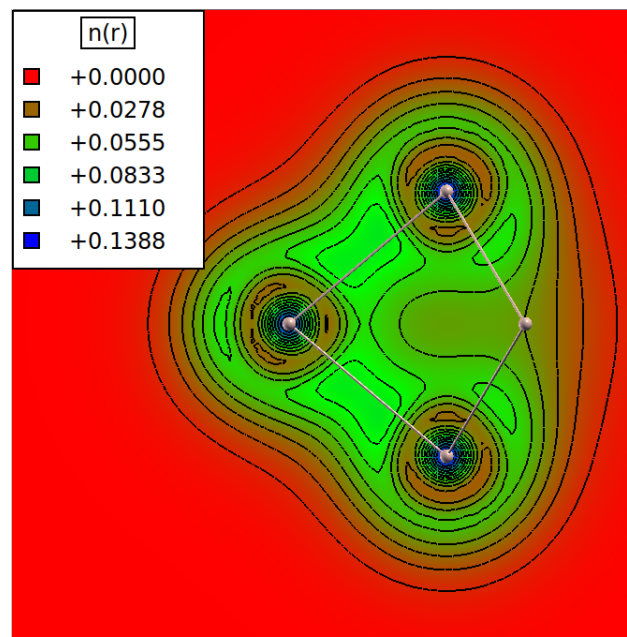
The lowest energy structure found in the previous small search will be further analysed thanks to the calculation of its electronic density. We implement again a 'SCF' simulation starting from the exact converged coordinates given by the third structure analysed before, since it appears to have the lowest energy after the recalibration of the parameters.

The visualization is visible in Figure 2. From these images it is visible that the structure enjoys the symmetry of a bipyramid with triangular basis. The first image 2a, that depicts

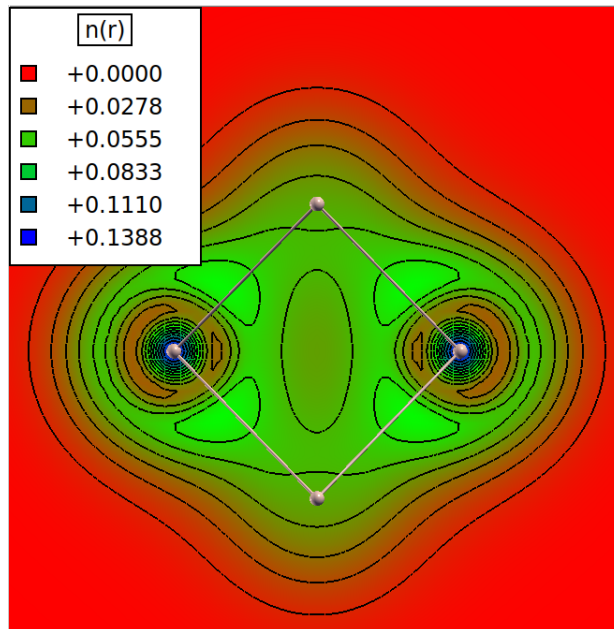
the view from above of one of the pyramids, emphasizes the isolines of the electron density in the plane defining the basis shared by the 2 pyramids. In this image we notice that electrons are not tightly gathered between the basis' atoms. The density is instead higher between the top/bottom extremity of the pyramids and the atoms in the basis. This is visible in Figure 2b, that portraits the side view of the structure. The plane in which the isolines are drawn is the one passing through the extremities and a basis point of the pyramids. We notice that this confirms the presence of bonds between these atoms. There is a lack of density also between the extremities themselves as show in the third image 2c, that emphasizes the view of the axis connecting the two top and bottom points.



(a) View of the triangular basis, from above



(b) View from the side of the bipyramid



(c) View at the axis of the bipyramid

Figure 2: Visualization of the electronic density thanks to XCrySden [4, 5]

3.4 Annealing Search

The annealing search has been implemented as described before and starting from the highest energy structure found in the previous sections, which was identified by the number 2, 'min2'. After applying the schedule and relaxing the obtained structure we reached the minimum associated with the name 'min5' in table 2 and in Figure 4e. Progressively, we would like to study in the same way this latter minimum, 'min5'. The first attempt was done applying the same annealing procedure as before but the result was not satisfying, since the cluster fell back to the same structure. Therefore, we tried to increase the initial temperature. In this new study, in fact, we changed the schedule to start from 3000 K and decrease the temperature by 135 K every 10 steps, for a total of 200 steps, reaching 300 K. Unfortunately this study brought us back to the original structure a second time.

Since the velocities are randomly initialized we performed another annealing starting again from the structure with highest energy, 'min2'. This time we reached a new minimum of which structure is completely different from what found up to now.

A recap of the energies and distances between the 10 unique atoms pairs is in Tables 4, 5 in the Appendix.

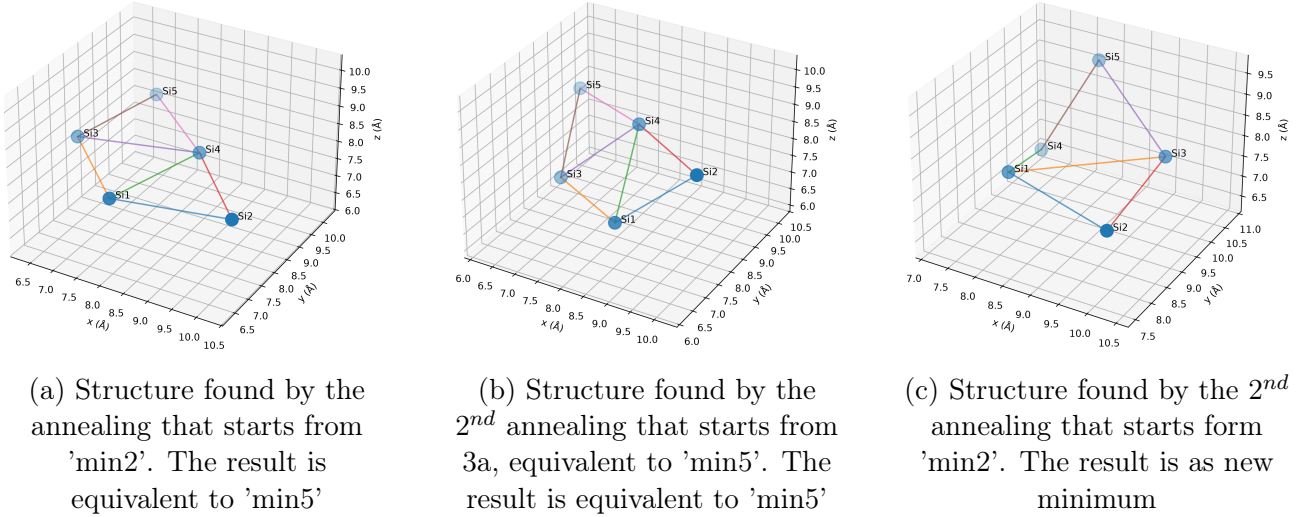


Figure 3: Results of the Simulated Annealing schedule

4 Conclusion and Comparison with Existing Data

This study was able to find 4 different stable configurations as minima of the energy surface in the space of possible configurations of this Si_5 cluster. The name given to them are 'min1', 'min2', 'min5' and 'min6'. Between them 'min1' is the lowest in energy of the structures found, followed by 'min6', 'min5' and finally 'min2'. The last two share also a planar shape and its no surprise that they are the highest in energy. The recap is in Tables 4 and 5 in the Appendix.

Several articles found that the Si_5 cluster takes the shape of a Trigonal Bipyramid [6, 7]. The structure with the lowest energy found in this project is in fact a Trigonal Bipyramid and the distances between the atoms, visible in Table 5, are compatible with the ones found in the literature: as stated in the article '*The small silicon clusters Si_n ($n=2-10$) and their anions: structures, thermochemistry, and electron affinities*' [6] "FOR NEUTRAL Si_5 , THE THREE EQUIVALENT 'BONDS' BETWEEN THE EQUATORIAL ATOMS ARE 3.137–3.198 Å, AND THE SIX EQUIVALENT AXIAL-EQUATORIAL BOND DISTANCES ARE 2.320–2.363 Å".

In the article '*Geometries and spectroscopic properties of silicon clusters (Si_5^+ , Si_5^- , Si_6 , Si_6^+ , and Si_6^-)*' [7], moreover, it is described a structure called 'edge-capped tetrahedron' which corresponds to the second lowest energy configuration found thanks to the Simulated Annealing schedule, 'min6'. Finally, we can state that the lowest energy structure found is coherent with the literature. Also, it is possible that some or all of the higher energy minima are spurious structures, not realistic but just a computational artifact.

5 Appendix

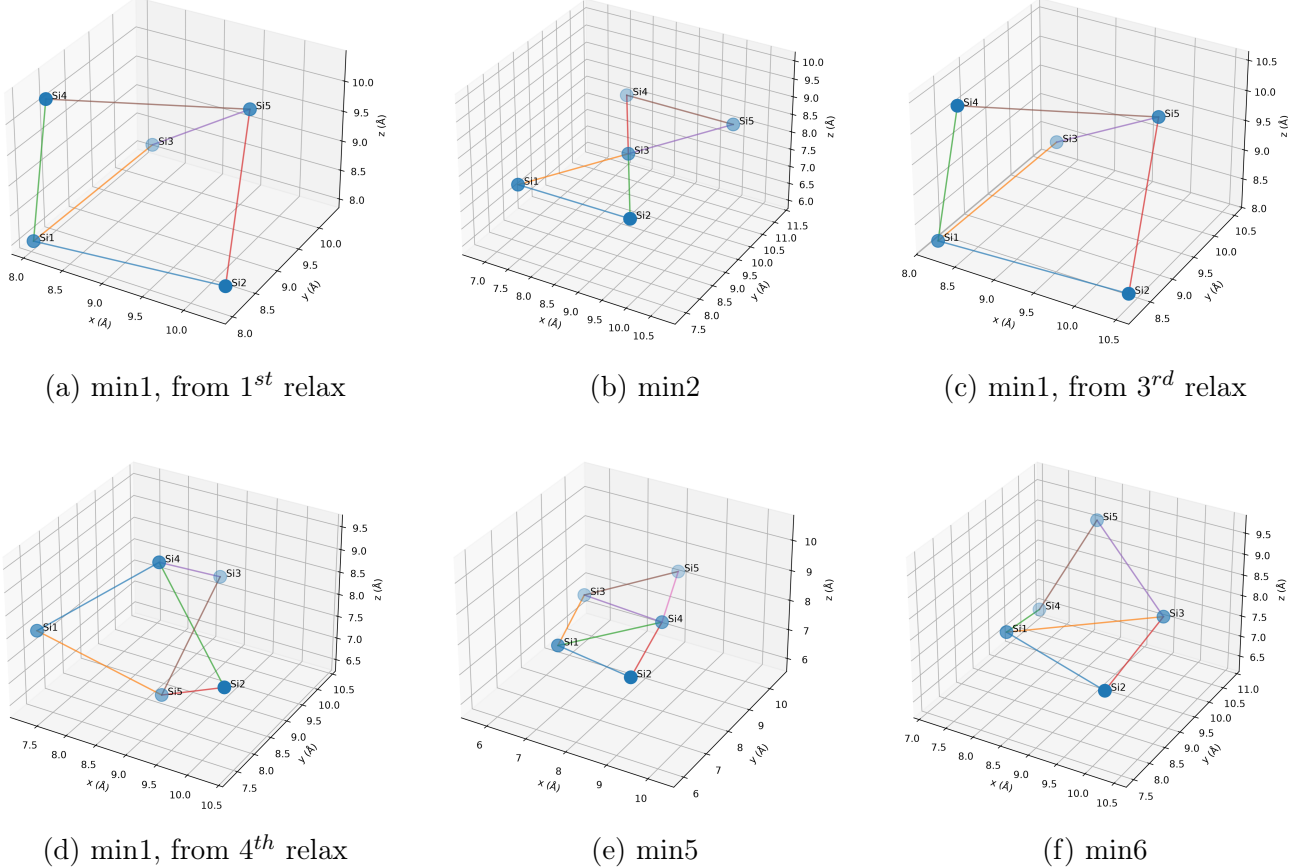


Figure 4: Results of the last relaxes after the convergence of 'ecutwfc' and the cell 'linear size'

Initial Structure	Minimum	Energy (Ry)	$\Delta E(Ry)$
1	min1	-50.6579275	0.0000002
2	min2	-50.4842374	0.1736903
3	min1	-50.6579277	0
4	min1	-50.6579253	0.0000024
5	min5	-50.5742100	0.0837177
3a	min5	-50.5741116	0.0838161
3b	min5	-50.5741264	0.0838013
3c	min6	-50.6046413	0.0532864

Table 4: Recap of minima found and their energies (last 3 are referring to the results of the Simulated Annealing in Figure 3)

I.S.	d_1 (Å)	d_2 (Å)	d_3 (Å)	d_4 (Å)	d_5 (Å)	d_6 (Å)	d_7 (Å)	d_8 (Å)	d_9 (Å)	d_{10} (Å)
1	2.313	2.313	2.313	2.313	2.313	2.313	2.977	3.066	3.066	3.066
2	2.271	2.271	2.271	2.271	2.324	2.324	3.902	3.902	4.542	4.542
3	2.313	2.313	2.313	2.313	2.313	2.313	2.976	3.066	3.066	3.066
4	2.313	2.313	2.313	2.313	2.313	2.313	2.977	3.066	3.067	3.067
5	2.254	2.254	2.319	2.332	2.333	2.402	2.402	4.025	4.026	4.796
3a	2.254	2.254	2.319	2.332	2.332	2.402	2.402	4.025	4.025	4.796
3b	2.254	2.254	2.319	2.332	2.333	2.402	2.402	4.025	4.026	4.796
3c	2.300	2.300	2.304	2.304	2.335	2.473	2.800	2.800	4.033	4.034

Table 5: Recap of minima found and the 10 atom-atom distances describing them (last 3 are referring to the results of the Simulated Annealing in Figure 3)

References

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